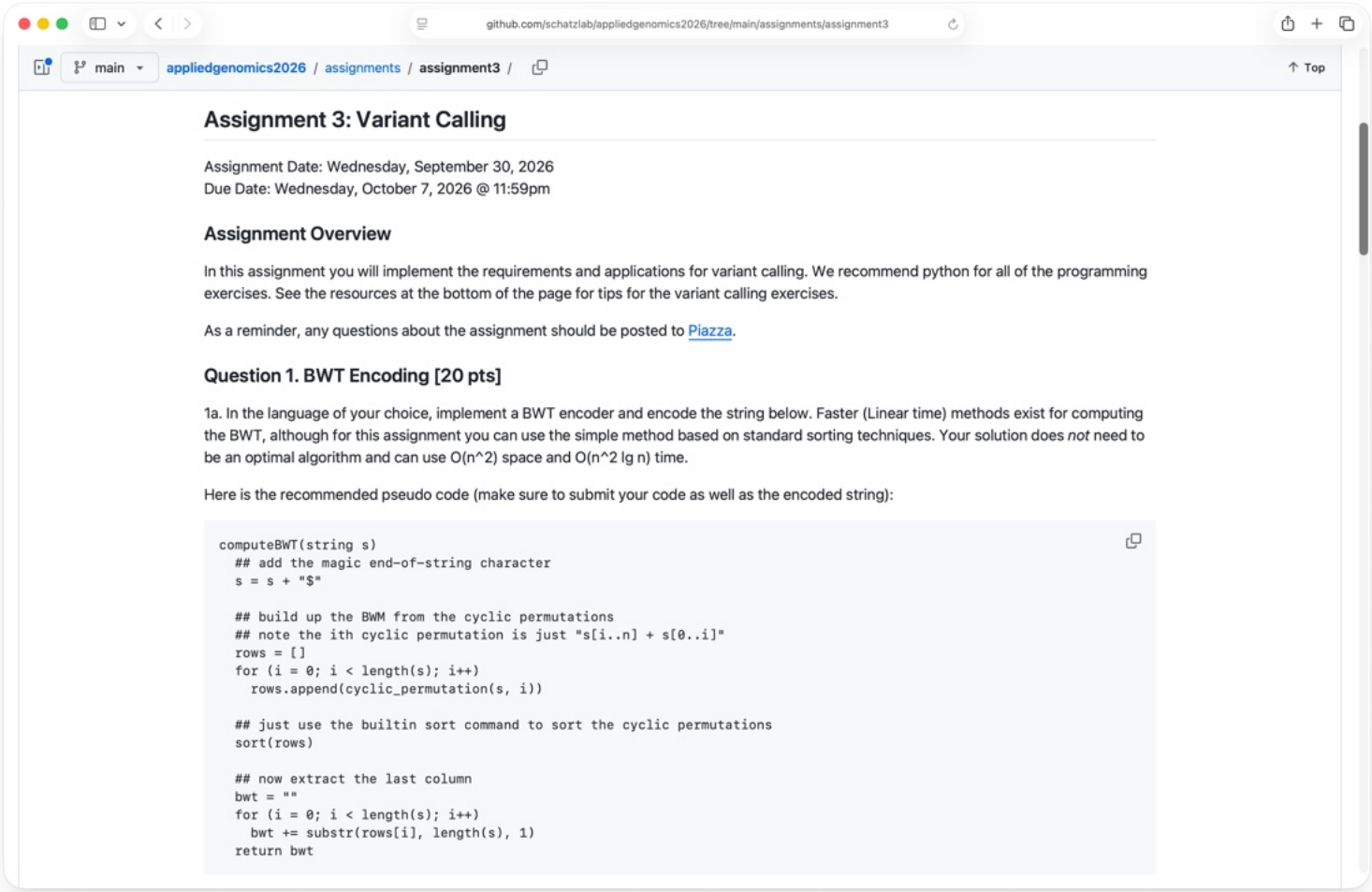
Review!

Michael Schatz
Oct 12, 2026

JHU EN.601.449/EN.601.649: Applied Comparative Genomics

Assignment 3 Due Wednesday Oct 7 by 11:59pm



Assignment 3: Variant Calling

Assignment Date: Wednesday, September 30, 2026
Due Date: Wednesday, October 7, 2026 @ 11:59pm

Assignment Overview

In this assignment you will implement the requirements and applications for variant calling. We recommend python for all of the programming exercises. See the resources at the bottom of the page for tips for the variant calling exercises.

As a reminder, any questions about the assignment should be posted to Piazza.

Question 1. BWT Encoding [20 pts]

1a. In the language of your choice, implement a BWT encoder and encode the string below. Faster (Linear time) methods exist for computing the BWT, although for this assignment you can use the simple method based on standard sorting techniques. Your solution does *not* need to be an optimal algorithm and can use O(n^2) space and O(n^2 lg n) time.

Here is the recommended pseudo code (make sure to submit your code as well as the encoded string):

```
computeBWT(string s)
  ## add the magic end-of-string character
  s = s + "$"

  ## build up the BWM from the cyclic permutations
  ## note the ith cyclic permutation is just "s[i..n] + s[0..i]"
  rows = []
  for (i = 0; i < length(s); i++)
    rows.append(cyclic_permutation(s, i))

  ## just use the builtin sort command to sort the cyclic permutations
  sort(rows)

  ## now extract the last column
  bwt = ""
  for (i = 0; i < length(s); i++)
    bwt += substr(rows[i], length(s), 1)
  return bwt
```

Project Proposal

Assignment Date: Monday October 12, 2026
Due Date: Wednesday, October 21 2026 @ 11:59pm

Review the Project Ideas page

Work solo or form a team for your class project of no more than 3 people.

The proposal should have the following components:

- Name of your team
- List of team members and email addresses
- Short title for your proposal
- 1 to 2 paragraph description of what you hope to do and how you will do it
- References to 3 to 4 relevant papers
- References/URLs to datasets that you will be studying (Note you can also use simulated data)
- Please add a note if you need me to sponsor you for a MARCC/skipjack account (high RAM, GPUs, many cores, etc)
- Please add a note if you need me to sponsor you for an AnVIL cloud billing account (dynamic resources)

Submit the proposal as a 1 to 2 page PDF on GradeScope (each team should submit one proposal and tag all people in the team). After submitting your proposal, I will provide feedback. If necessary, we can schedule a time to discuss your proposal, especially to ensure you have access to the data that you need. The sooner that you submit your proposal, the sooner we can schedule the meeting.

Later, you will present your project in class during the last week of class. You will also submit a written report (5-7 pages) of your project, formatting as a Bioinformatics article (Intro, Methods, Results, Discussion, References).

Please use Piazza to coordinate proposal plans!

Exam 1 Wednesday

12	10/12/26	M	Review 1	Project Proposal	
13	10/14/26	W	Exam 1 (In class)		



Exam 1 Review

What to study?

Closed book / no electronics in class exam [75 min]:

- One sheet of paper for notes (double sided)
- Recommend powerpoint / gslides / keynote but hand-written is okay
- Bring a pen / pencil!

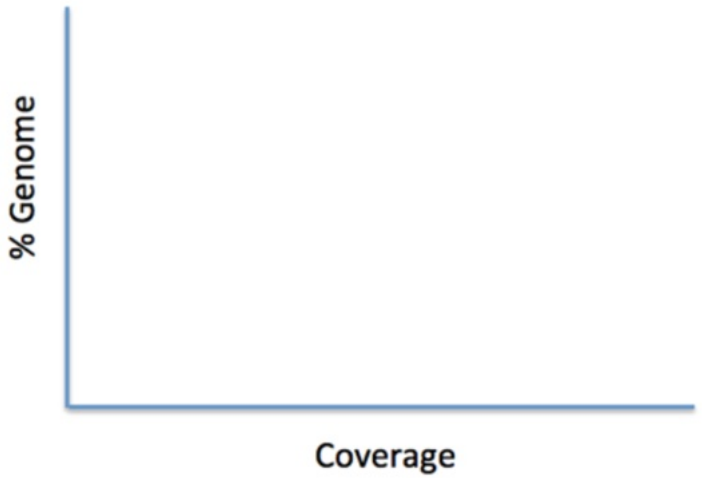
1. **Homework assignments**
   - Make sure you understand the key concepts, algorithms, data structures, and methods involved
   - Don't worry about running software or writing code, but you should know what are the major software tools and how they work

2. **Lectures [with videos on canvas!]**
   - Make sure you understand the key concepts, algorithms, data structures, and methods involved :)
   - No "complicated" numerical computing, but you should be able to sketch a distribution, approximate values, & other simple calculations

3. **Readings**
   - These are to provide more details and provide additional perspectives
   - You are not responsible for plots/content only presented in the readings
   - Of course, you are free to read/watch/discuss additional resources

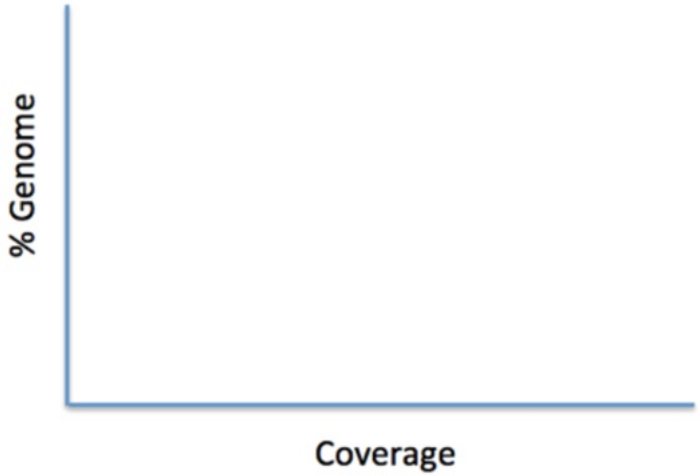
Sample Question

Q3. The Maryland blue crab genome is 1 Gbp in size. Approximately how many 100bp reads should we sequence so that we expect at least 99.85% of the genome will be sequenced at least 40 times? Sketch the expected coverage distribution for this number of reads; be sure to clearly label the mean coverage, and how 40 fold coverage relates to the mean. (Hint: In a normal distribution, 68.2% of the data is within 1 standard deviation of the mean, 95.4% within 2, 99.7% within 3, and 99.9% within 4)



Sample Question

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Also expect a few multiple choice, short answer, and longer essay questions

Exam Topics

Genomics

- Genomics Technologies
  - Illumina, PacBio, Nanopore
- Coverage, Kmers
- Genome Assembly
- Repeats, Heterozygosity
- Whole Genome Alignment
- Dotplots
- Jaccard Coefficient
- Average Nucleotide Identity
- Read mapping
- Variant Identification

Quantitative Techniques

- Normal, Poisson, Binomial
- P-value
- de Bruijn and overlap graphs
- Quality Values (Phred Scale)
- Full text indexing
- Hashing & Suffix Arrays
- Seed & Extend
- Edit Distance
- PCA / t-SNE / UMAP
- Neural networks
- Convolutional Neural Networks

What is the goal? What is the approach? What are the key challenges?

How did we explore these topics in the homeworks and lectures?